Lorem Ipsum

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BTP report submitted in partial fulfilment of the requirements
for the Degree of B.Tech. in Computer Science & Engineering

on October 27, 2026

BTP Track: Research

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Student Declaration

I hereby declare that the work presented in the report entitled **"Lorem Ipsum"** submitted by me for the partial fulfilment of the requirements for the degree of *B.Tech. in Computer Science & Engineering* at Indraprastha Institute of Information Technology, Delhi, is an authentic record of my work carried out under the guidance of **Prof. Lorem Ipsum** . Due acknowledgements have been given in the report for all material used. This work has not been submitted elsewhere for the reward of any other degree.

Lorem Ipsum **Place & Date:** October 27, 2026

Certificate

This is to certify that the above statement made by the candidate is correct to the best of my knowledge.

Prof. Lorem Ipsum **Place & Date:** October 27, 2026

Abstract

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Keywords: (e.g.Mobile computing, security, complexity, algorithms, image analysis, machine learning, information retrieval)

Acknowledgment

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Work Distribution

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Contents

Student Declaration i

Abstract ii

Acknowledgment iii

Table of Contents v

1 Introduction 1

1.1 Background . . . 1
1.2 Digital Twins in Medicine and Nutrition . . . 2
1.3 Computational Gastronomy and Recipe Data . . . 2
1.4 Problem Statement and Research Gap . . . 3
1.5 Research Question and Objectives . . . 3
1.6 Approach Overview . . . 4
1.7 Contributions . . . 4
1.8 Scope and Limitations . . . 5
1.9 Thesis Outline . . . 5

2 Literature Review 6

2.1 MASLD and Its Dietary Management . . . 6
2.1.1 Definition and burden . . . 6
2.1.2 Evidence for lifestyle and dietary treatment . . . 6
2.2 Non-Invasive Assessment of Liver Fat . . . 7
2.3 Machine-Learning Prediction of Steatosis and Liver Fat . . . 8
2.3.1 Classification of steatosis in NHANES . . . 8
2.3.2 Prediction of liver fat as a continuous value . . . 9
2.3.3 Why prediction from routine data is bounded . . . 10
2.4 Dietary Assessment and Personalised Nutrition . . . 10
2.5 Digital Twins in Health and Nutrition . . . 11
2.5.1 Definitions and requirements . . . 11

2.5.2 Metabolic and nutrition twins . . . 12
2.5.3 From prediction to "what if" . . . 12
2.6 Computational Gastronomy and Food Recommendation . . . 13
2.6.1 Recipe and food resources . . . 13
2.6.2 Food recommender systems . . . 13
2.6.3 Recipe modification and adaptation . . . 14
2.6.4 Menu planning, diversity and additional criteria . . . 14
2.7 Evaluating Digital Twins Without Longitudinal Data . . . 15
2.8 Summary and Research Gap . . . 16

3 Methodology **17**

4 Results **18**

Chapter 1

Introduction

1.1 Background

Metabolic dysfunction-associated steatotic liver disease (MASLD), formerly non-alcoholic fatty liver disease (NAFLD), is the most common chronic liver disease worldwide. It affects about 30
prevalence has risen steadily over the past three decades [**younossi2023global**, **riazi2022prevalence**]. In 2023 a multisociety consensus renamed the condition to reflect its metabolic origin: steatotic liver disease accompanied by at least one cardiometabolic risk factor, such as excess weight, raised blood glucose or dyslipidaemia [**rinella2023nomenclature**]. Untreated, MASLD can progress from simple fat accumulation (steatosis) to inflammation, fibrosis and cirrhosis, and it is closely linked to type$\tilde{2}$ diabetes and cardiovascular disease. Diet is central to its management. No dietary pattern is a cure, but the current European guideline recommends lifestyle change, centred on diet quality and weight loss, as the first-line treatment for most patients [**easl2024masld**]. The evidence is graded by the amount of weight lost: in a prospective study with paired liver biopsies, a loss of at least 5
weight reduced steatosis, 7–10
to resolution of steatohepatitis in most patients and to regression of fibrosis in many [**vilargomez2015weight**]. Beyond weight, the guideline advises a Mediterranean-style diet, fewer foods rich in saturated fat and sugars, and the avoidance of sugar-sweetened beverages [**easl2024masld**]. Turning this guidance into daily meals is where patients struggle. Guidelines speak in nutrients and dietary patterns (``limit saturated fat", ``increase fibre"), whereas people decide what to eat one recipe at a time, constrained by the cuisine they know, the ingredients they have, the time they can spend cooking and what they enjoy. The same advice also does not suit every patient equally: the priority for someone whose diet is already high in fibre but rich in saturated fat differs from that of someone with the opposite pattern. Bridging the gap between population-level evidence and an individual patient's plate is the problem this thesis addresses. Liver fat itself can be measured non-invasively. Vibration-controlled transient elastography (FibroScan) reports the controlled attenuation parameter (CAP), which grades steatosis with established cut-offs [**karlas2017cap**]. The US National Health and Nutrition Examination Survey (NHANES) performed this examination on a nationally representative sample in its 2017–March 2020 and 2021–2023 cycles, together with blood tests, body

measurements, questionnaires and a 24-hour dietary recall [**nhanes˙lux**]. In the 15,300 participants with a complete examination used in this work, 52
for any steatosis.

1.2 Digital Twins in Medicine and Nutrition

A digital twin is a virtual representation of a specific physical system that is updated with data from that system, can predict how it will behave, and is used to inform decisions about it [**nasem2024digitaltwins**]. In medicine the physical system is a patient, or one of their organs. Medical digital twins are most advanced in cardiology and oncology, and their development faces common challenges in data integration, validation and privacy [**laubenbacher2024medicine**, **katsoulakis2024digital**]. What distinguishes a twin from a one-off prediction model is that it is specific to one patient, is updated as new data arrive, and can simulate how that patient's state may change under different interventions [**nasem2024digitaltwins**, **katsoulakis2024digital**]. Nutrition is a natural application. People respond differently to the same food [**zeevi2015personalized**], and the ``virtual digital twin" concept in precision nutrition proposes combining a person's clinical, lifestyle and dietary data into a model that can test dietary choices before they are made [**gkouskou2020virtual**]. Early clinical results are encouraging: in a one-year randomised trial in people with type$\tilde{2}$ diabetes, nutrition guided by a digital twin that predicted each person's post-meal glucose response improved markers of liver fat [**joshi2023digitaltwin**]. In MASLD specifically, the European ARTEMIS project is building virtual twins from retrospective data on several thousand patients [**artemis2024**]. None of these systems, however, evaluates complete, existing recipes against a patient's evidence-based dietary needs.

1.3 Computational Gastronomy and Recipe Data

Computational gastronomy treats food and cooking as data that can be analysed and reasoned about [**bagler2024gastronomy**]. Structured resources now make this possible at scale. RecipeDB contains more than 118,000 recipes from 74 countries, each with its ingredients, cooking processes, utensils, cuisine and nutritional profile [**batra2020recipedb**]; FlavorDB describes the flavour molecules found in foods [**garg2018flavordb**]; and literature-mined resources link culinary herbs and spices to health outcomes [**rakhi2018spices**]. Related work in menu planning uses multi-objective optimisation to compose daily menus that meet nutritional targets [**sahin2024menu**]. These resources are designed mainly for exploring and querying culinary data, or for meeting generic nutrient targets. They do not, by themselves, decide which existing recipes suit a particular patient, given that patient's current liver state, the strength of the clinical evidence behind each dietary change, and the practical realities of their kitchen.

1.4 Problem Statement and Research Gap

Three bodies of work hold complementary pieces of the problem. Clinical evidence links dietary changes to liver outcomes; patient-specific models, including digital twins, give an estimate for an individual's health state; and recipe databases describe the food people actually cook. These components are usually studied separately. In particular, we found no framework that evaluates and ranks existing recipes for an individual patient by combining

- ****health relevance****, derived from the patient's current liver and metabolic state and from graded intervention evidence;
- ****cuisine compatibility****, reflecting the patient's culinary context; and
- ****practical feasibility****, reflecting cooking time, number of ingredients and complexity,

within a transparent framework in which every recommendation can be traced back to the evidence that supports it.

1.5 Research Question and Objectives

This thesis asks:

> **Can a patient-specific liver digital twin, built from clinical data and* *graded intervention evidence, support the evidence-grounded evaluation and* *ranking of existing recipes according to their health relevance, cuisine* *compatibility and cooking feasibility for an individual patient?**

To answer it, the work pursues five objectives:

1. Estimate each patient's current liver state, as calibrated probabilities of fatty liver, from routinely collected clinical data.
2. Translate that state, together with the patient's own reported diet, into evidence-graded nutritional priorities, without inventing individual numeric targets.
3. Evaluate and rank existing recipes from RecipeDB against those priorities, the patient's cuisine and practical constraints.
4. Compare recipe choices as scenarios whose projected effects are grounded in clinical evidence, and update the patient's state as new data arrive.
5. Evaluate each component against appropriate baselines and state clearly what has and has not been validated.

1.6 Approach Overview

The proposed system, HepatoTwin, is organised in seven layers (Figure~**??**fig:hepatotwin$_o verview$).$Layer\tilde{1}holdsthe$
$demographics, bodymeasurements, bloodtests, questionnaireanswersanda24-hourdietaryrecall.Layer\tilde{2}, thecore$
$-outcomefindingswithexplicitevidence-strengthlabels, andisadjustedbythepatient'sownintake.Layer\tilde{4}, therecip$
$itscorescandidaterecipesforhealthrelevance, cuisinecompatibility, practicalfeasibilityandacceptability, ranksfi$
$linkedreferencevalue, andexplainseachrecommendation.Layer\tilde{5}essentiallycomparesrecipescenariosandprojects$
$dimensionalliverandanexplainabledashboardareusedinLayers\tilde{6}and\tilde{7}forvisualisation.Thetwinupdatesitsstatean$
$statemodelisintendedforpatientsaged12to59andisevaluatedagainstreliableFibroScanexaminations; ages60ando$

1.7 Contributions

The main contributions of this thesis are:

1. ****A calibrated liver-state model from routine data.**** Trained on both NHANES FibroScan cycles, the model detects steatosis with ROC-AUC 0.881 (CAP $\geq$ 248$\tilde{\text{dB/m}}$) and 0.879 (CAP $\geq$ 280$\tilde{\text{dB/m}}$) on held-out patients within its scope, above the Fatty Liver Index (0.861) and the Hepatic Steatosis Index (0.836) on the same patients [**bedogni2006fli**, **lee2010hsi**], and holds its performance on a later survey cycle. It also predicts CAP with a held-out R^2 of 0.56, comparable to or above published models built on routine data.

2. ****An evidence engine linking patient state to graded dietary** **xA0; priorities.**** Every priority cites named trials, meta-analyses or guidelines with an explicit strength label, observational findings are recorded but never used as effects, and the patient's own 24-hour recall personalises the priorities.

3. ****NutriTwin, an evidence-gated multi-criteria recipe ranking** **xA0; engine.**** On 300 held-out patients, 83
   every evidence-linked reference value, the patient's cuisine and the practical limits at once, against 73
   the health gate and under 2
   only method whose recommendations always met the nutritional references.

4. ****Scenario comparison and weekly meal plans**** that respect the patient's cuisine and nutritional references, with projected effects labelled by the strength of their evidence.

5. ****An open, modular digital-twin framework****, demonstrated for MASLD, together with an in-silico evaluation design that tests how the twin updates on virtual patients before real follow-up data exist.

1.8 Scope and Limitations

NHANES is cross-sectional: each participant was examined once. The twin's estimate of the current state is therefore validated on held-out patients, but its projections of future states cannot be compared with real repeat measurements. They are presented as simulated projections grounded in clinical evidence, never as clinical forecasts, and the twin's updating is evaluated on virtual patients. For validation with real longitudinal data, deployment with patients who enter their own measurements over time is required. The recommendations are research outputs: they support, and do not replace, clinical care.

1.9 Thesis Outline

Chapter$\tilde{2}$ reviews related work on MASLD and its dietary treatment, non-invasive liver fat assessment, machine-learning prediction of steatosis, dietary assessment, digital twins, computational gastronomy and the evaluation of twins without longitudinal data. Chapter$\tilde{3}$ describes the methodology: the NHANES cohort and the recipe corpus, the liver-state model, the evidence engine, the recipe evaluation and ranking engine, and the scenario and updating components of the twin. Chapter$\tilde{4}$ reports the experiments carried out to improve the prediction of CAP, and explains why its performance reaches a plateau. Chapter$\tilde{5}$ presents the results of each component against its baselines.

Chapter 2

Literature Review

This chapter reviews the work on which HepatoTwin builds and the gaps it addresses. It follows the structure of the system: the clinical problem and the evidence for dietary treatment (Section 2.1); how liver fat is measured without biopsy (Section 2.2); machine-learning models that estimate steatosis from routine data, and the limits on how well they can do so (Section 2.3); how diet is measured and personalised (Section 2.4); digital twins in health and nutrition (Section 2.5); computational gastronomy and food recommendation (Section 2.6); and how a twin can be evaluated when no longitudinal data are available (Section 2.7). Section 2.8 summarises the gap.

2.1 MASLD and Its Dietary Management

2.1.1 Definition and burden

Metabolic dysfunction-associated steatotic liver disease (MASLD) is defined as hepatic steatosis together with at least one of five cardiometabolic risk factors, in the absence of another cause [**rinella2023nomenclature**]. The 2023 multi-society nomenclature replaced non-alcoholic fatty liver disease (NAFLD), removed the reliance on exclusion and stigmatising terms, and introduced MetALD for patients with MASLD who also drink moderate amounts of alcohol. A meta-analysis of studies using imaging to diagnose NAFLD estimated a global adult prevalence of 30.05%, rising from 25.26% in 1990–2006 to 38.00% in 2016–2019 [**younossi2023global**]; a second meta-analysis reached a similar overall estimate [**riazi2022prevalence**]. Most patients have simple steatosis, but a minority progress to steatohepatitis, fibrosis and cirrhosis, and the disease is closely associated with type 2 diabetes and cardiovascular risk [**easl2024masld**, **rinella2023aasld**].

2.1.2 Evidence for lifestyle and dietary treatment

Both the European (EASL-EASD-EASO) and American (AASLD) guidelines name lifestyle change as the first-line treatment for most patients [**easl2024masld**, **rinella2023aasld**]. The strongest evidence is about weight loss. In a prospective study of 293 patients with paired liver biopsies,

weight loss through lifestyle change produced dose-dependent histological improvement: losses of 5% or more improved steatosis and inflammation, and losses of 10% or more resolved steatohepatitis in 90% of patients and led to fibrosis regression in 45% [**vilargomez2015weight**]. A meta-analysis of 22 randomised trials with 2,588 participants found that weight-loss interventions lowered ALT by 9.81 U/L and improved liver fat and histological activity, but found no significant change in fibrosis [**koutoukidis2019weightloss**]. The thresholds of 5%, 7–10% and 10% weight loss are now written into the European guideline [**easl2024masld**]. Liver fat also responds quickly: in people with type 2 diabetes on an 800 kcal per day diet, hepatic triglyceride fell by 30% after the first week, and eventually dropped from 12.8% to 2.9% over eight weeks [**lim2011reversal**].

The evidence for specific dietary components is more general, and this variation is important for any system that turns evidence into advice. A 2026 meta-analysis of randomised trials, which graded the certainty of each finding with GRADE, found that a Mediterranean-style diet reduced MRI-PDFF by 1.37 percentage points and improved liver stiffness, while omega-3 supplementation and low-carbohydrate high-fat diets did not change ALT [**stern2026dietary**]. Dietary interventions including calorie restriction improved liver enzymes, steatosis and stiffness in another meta-analysis [**haigh2022mediterranean**]. Yet a meta-analysis of ten trials found that the Mediterranean diet reduced AST and GGT but not ALT [**sangouni2022mediterranean**], and it was no better than a low-fat diet in the short term [**xiong2024mediterranean**].

Feeding trials show that the type of surplus energy matters. When 38 overweight adults were overfed by 1,000 kcal per day for three weeks, liver fat rose by 55% with saturated fat, 33% with simple sugars and 15% with unsaturated fat [**luukkonen2018saturated**]. A meta-analysis of 13 trials with 260 participants found that fructose raised liver fat and ALT when it added surplus energy, but had no effect when it replaced other carbohydrate in the same amount of energy [**chiu2014fructose**]. Exercise also reduces liver fat: in a meta-analysis of trials measuring liver fat by MRI, exercise was associated with a relative reduction of 30% or more, the threshold that predicts histological response, even without clinically significant weight loss [**stine2023exercise**].

Two results follow. The available evidence varies in strength, with biopsy-based dose–response findings on weight loss generally providing more direct support than meta-analyses based on liver enzyme changes. The literature also includes studies reporting no significant effects alongside those showing positive outcomes. Food recommendation systems should therefore communicate the quality and certainty of the supporting evidence for each recommendation instead of assigning equal weight to all findings. Second, the evidence is expressed in terms of energy, weight change, nutrients and dietary patterns, never single recipes; any mapping from evidence to a recipe is an inference that must be made explicit.

2.2 Non-Invasive Assessment of Liver Fat

Liver biopsy is the reference standard for grading steatosis and fibrosis but is invasive and impractical for screening. Magnetic resonance imaging proton-density fat fraction (MRI-PDFF) and proton spectroscopy (^{1}H-MRS) measure liver fat accurately but are costly. Vibration-controlled transient

elastography (FibroScan) is cheaper and widely used: its controlled attenuation parameter (CAP), reported from 100 to 400 dB/m, grades steatosis, with cut-offs of 248, 268 and 280 dB/m for grades S1, S2 and S3 derived from an individual patient data meta-analysis [**karlas2017cap**].

CAP is, however, an imperfect measure of liver fat. A review of the technique reports that agreement between two operators examining the same patient has a concordance correlation of 0.82, falling to 0.65 in patients with BMI of 30 kg/m^2 or more and to 0.44 when CAP is below 240 dB/m [**ferraioli2021cap**]. The same review notes that CAP accuracy falls when the interquartile range (IQR) of the ten acquisitions exceeds 40 dB/m, which is why that limit is used as a quality criterion. In adults with NAFLD, CAP correlated with MRI-PDFF at $r = 0.66$, falling to $r = 0.37$ for changes over time [**ahn2018cap**]. Part of every CAP reading is therefore measurement variation, and the variation is largest in exactly the people who are hardest to examine. This limits how closely any model can predict CAP (Section 2.3.3).

Simple clinical scores offer a cheaper alternative. The Fatty Liver Index (FLI) combines BMI, waist circumference, triglycerides and GGT [**bedogni2006fli**]; the Hepatic Steatosis Index (HSI) combines the ALT/AST ratio, BMI, sex and diabetes [**lee2010hsi**]; and the NAFLD liver fat score combines metabolic syndrome, type 2 diabetes, fasting insulin, AST and the AST/ALT ratio, reaching an AUC of 0.86–0.87 against spectroscopy, with genetic information adding less than 1% [**kotronen2009prediction**]. These scores are the natural baselines for any new steatosis model.

2.3 Machine-Learning Prediction of Steatosis and Liver Fat

2.3.1 Classification of steatosis in NHANES

The US National Health and Nutrition Examination Survey (NHANES) added FibroScan to its 2017–March 2020 cycle, and several studies have since trained classifiers to detect CAP-defined steatosis from routine data (Table 2.1). Gradient-boosted trees, usually XGBoost [**chen2016xgboost**], perform best or near best in most of them, and the strongest predictors are consistently body-size measures (waist circumference and BMI), followed by liver enzymes, glucose and insulin markers, and triglycerides. Model explanations, typically SHAP values [**lundberg2017shap**], are widely reported.

Du et al. used 6,814 participants and found logistic regression, XGBoost and LightGBM comparable (AUC 0.815–0.842), so that the added complexity of boosting bought little [**du2026masld**]. Liu et al. compared models on an automated machine-learning platform in 4,177 participants; XGBoost reached an AUC of 0.859, and the authors imputed missing values by multiple imputation and reported calibration curves and decision-curve analysis [**liu2024automl**]. Zhang et al. reported an AUC of 0.874 on 2,460 complete cases [**zhang2025masld**]. Boullion et al. identified biomarkers of steatosis and stiffness; propensity matching on age and sex required complete data and reduced the steatosis sample to 2,714, so their models were trained on the larger unmatched data, where random forests reached 79.6% accuracy [**boullion2025biomarker**]. Zhu et al. trained a three-grade model on 215,145 non-obese Chinese adults, reaching a macro-average AUC of 0.941 internally and 0.874

in 726 non-Hispanic Asian NHANES participants with complete data [**zhu2026nonobese**]; this is one of the few studies with external validation, and it shows the usual fall in performance when a model moves to a new population.

Table 2.1: Machine-learning classifiers of CAP-defined steatosis using NHANES.

Study	Population	Outcome	Best model	Performance
Du et al. (2026) [**du2026masld**]	6,814 adults, 2017–2020	MASLD, CAP $\geq$ 280	LR, XGBoost, LightGBM	AUC 0.815–0.842
Liu et al. (2024) [**liu2024automl**]	4,177, fasting subsample; imputed	CAP $\geq$ 302	XGBoost (AutoML)	AUC 0.859; calibration curve
Zhang et al. (2025) [**zhang2025masld**]	2,460 complete cases	CAP-defined MASLD	XGBoost	AUC 0.874
Boullion et al. (2025) [**boullion2025biomarker**]	Unmatched sample (matched subset 2,714)	CAP $\geq$ 238	Random forest	Accuracy 79.6%
Zhu et al. (2026) [**zhu2026nonobese**]	215,145 non-obese Chinese adults; 726 NHANES Asians	CAP grades (S0, S1, S2–S3)	XGBoost	Macro AUC 0.941 internal; 0.874 external

Three limitations recur. Most studies use only the 2017–2020 cycle, and most discard participants with any missing value. Calibration is reported only occasionally [**liu2024automl**], although a risk model whose output will be shown to patients must produce probabilities that mean what they say; isotonic regression is a standard way to obtain them from tree ensembles [**zadrozny2002calibration**]. And all are single snapshots, used to classify a participant once rather than to track or support them over time.

2.3.2 Prediction of liver fat as a continuous value

Fewer studies predict the amount of liver fat, and their results are more sobering (Table 2.2). In 28,796 UK Biobank adults, the Fatty Liver Index explained 28% of the variance in MRI-PDFF (33–37% across cross-validation folds); an extended model with twelve variables (FLI+) explained 54% on average across four cross-validation folds (0.520–0.557) [**asaturyan2022fli**]. A UK Biobank preprint reported R^2 of 0.37 from anthropometric and blood measures in 40,533 participants with MRI-PDFF and 0.34 from plasma metabolites in 9,723, against 0.63–0.64 from DXA body-composition images analysed by a convolutional network [**insitro2024ukb**]. Haas et al., also in UK Biobank, found that BMI correlated with MRI liver fat at $r = 0.42$ and that adding 24 clinical factors and biomarkers raised the correlation only to $r = 0.58$ ($R^2 \approx 0.34$), concluding that liver fat cannot be reliably estimated from clinical factors [**haas2021liverfat**]. In Finnish adults measured by spectroscopy, 15–25% of the variance could be explained by liver enzymes alone; an equation including a specialist marker (phosphorylated IGFBP-1) reached an adjusted R^2 of 0.49 in validation and an AUROC of 0.84 for NAFLD, against 0.72 for FLI and 0.62 for HSI [**petaja2016igfbp**]. Higher values come from small, selected samples: in 272 patients of a Chinese obesity clinic, a four-variable equation predicted CAP with $R^2 = 0.76$ in derivation and 0.65 in validation [**liu2022surrogate**], and a four-variable index correlated with MRI-PDFF at $\rho = 0.74$ in 149 overweight or obese adults, validated only by bootstrap resampling [**chen2023shsi**]. No

study reporting R^2 for CAP in NHANES was found.

Table 2.2: Prediction of liver fat as a continuous value.

Study	Population	Reference	Inputs	R^2
UK Biobank preprint (2024) [**insitro2024ukb**]	40,533; 9,723; about 46,000 UK adults	MRI-PDFF	Body and blood measures; metabolites; DXA images	0.37; 0.34; 0.63–0.64
Asaturyan et al. (2022) [**asaturyan2022fli**]	28,796 UK adults	MRI-PDFF	FLI; FLI with 12 variables (FLI+)	0.28–0.36; 0.54 (4-fold mean)
Haas et al. (2021) [**haas2021liverfat**]	UK adults, 1,214 test	MRI	BMI; BMI with 24 clinical factors	r 0.42; r 0.58
Petäjä et al. (2016) [**petaja2016igfbp**]	378 Finnish adults	^{1}H-MRS	Enzymes only; equation with phosphorylated IGFBP-1	0.15–0.25; 0.49 (adjusted)
Liu et al. (2022) [**liu2022surrogate**]	272, obesity clinic	CAP	BMI, neck circumference, TG, C-peptide	0.76 / 0.65
Chen et al. (2023) [**chen2023shsi**]	149, overweight or obese	MRI-PDFF	ALT, AST, insulin, 1-h glucose	$\rho = 0.74$
Kotronen et al. (2009) [**kotronen2009prediction**]	470 Finnish adults	^{1}H-MRS	Metabolic and liver markers	Not reported

2.3.3 Why prediction from routine data is bounded

Two findings explain why continuous liver fat is hard to predict from routine measurements. First, liver fat is strongly heritable: in a prospective study of monozygotic and dizygotic twins using MRI-PDFF, the heritability of steatosis was 0.52 after adjustment for age, sex and ethnicity [**loomba2015heritability**], and in UK Biobank measured genetic variants together explained up to 30% of the variance in liver fat [**haas2021liverfat**]. Variants such as those in PNPLA3 change liver fat at the same body size, and routine clinical data contain none of this information. Second, the reference measurement itself is noisy (Section 2.2): a model can explain no more of the variation in a single CAP reading than the reading shares with the patient's true liver state. Together, these findings imply that a model built on routine data should reach a moderate R^2 for liver fat, between about 0.3 and 0.55 in the large studies above, and good discrimination of steatosis, which is the pattern seen in Tables 2.1 and 2.2. For a digital twin, the implication is that a calibrated probability of steatosis is a more defensible state variable than a single predicted CAP value.

2.4 Dietary Assessment and Personalised Nutrition

Personalising dietary advice requires knowing what a person usually eats. NHANES collects one or two 24-hour dietary recalls per participant. A single recall describes the average intake of a group well, but an individual's usual intake poorly, because intake varies from day to day. Statistical methods developed at the US National Cancer Institute estimate usual intake from two or more recalls by separating within-person from between-person variation [**dodd2006usual**]. For foods that are not

eaten every day, the NCI method uses a two-part model that separates the probability of eating a food on a given day from the amount eaten on days it is eaten, allows the two parts to be correlated, and can add covariates such as age, sex or a food frequency questionnaire; in simulation it estimated the distribution of usual intake substantially better than earlier methods [**tooze2006usual**]. Self-reported intake is also biased: in the OPEN study, comparison with recovery biomarkers showed substantial under-reporting, particularly of energy [**subar2003open**]. Diet quality can be summarised with the Healthy Eating Index-2020, a 0–100 score of 13 components aligned with the 2020–2025 Dietary Guidelines for Americans; its components are unchanged from HEI-2015 for ages 2 and over, and a separate index was introduced for toddlers [**shamswhite2023hei**]. Computing it requires food-group data from the USDA Food Patterns Equivalents Database.

Personalisation is motivated by individual variation in response. Zeevi et al. showed in 800 people that post-meal glucose responses to identical meals differ widely and can be predicted from personal data, and that diets personalised on these predictions lowered post-meal glucose [**zeevi2015personalized**]. For a liver twin, these findings suggest two requirements: rather than single-day intake, dietary inputs should represent the usual intake, and the system should become more personal as more of the patient's own data arrive.

2.5 Digital Twins in Health and Nutrition

2.5.1 Definitions and requirements

The US National Academies define a digital twin as a set of virtual information constructs that mimics the structure, context and behaviour of a physical system, is dynamically updated with data from it, has a predictive capability and informs decisions; the bidirectional flow between the virtual and physical system is central, and credibility rests on verification, validation and uncertainty quantification (VVUQ) [**nasem2024digitaltwins**]. A scoping review of health twins frames them as a physical entity, its virtual representation and the connection between them, and found that the term is used for systems ranging from static models to fully coupled ones [**katsoulakis2024digital**]. A perspective on medical twins describes them as most advanced in cardiology and oncology, and names data integration, validation and privacy as the main obstacles [**laubenbacher2024medicine**].

Two architectural proposals are relevant to a modular twin. Nitschke et al. propose a general patient twin built on a bipartite knowledge graph that links patient attributes to existing predictive models; fusion models based on ensemble learning combine the models' outputs, and a propagation scheme passes information through the graph each time new patient data arrive. They describe the design as modular, evolving, informed, interpretable and explainable [**nitschke2026design**]. Nye structures patient data as a knowledge graph built on medical ontologies such as SNOMED CT and proposes closed-form continuous-time ("liquid") neural networks to model the patient's evolving state [**nye2023knowledge**]. Both are conceptual designs without empirical evaluation, but they share a principle that HepatoTwin adopts: separate modules, each with a defined role, connected by an explicit patient state.

2.5.2 Metabolic and nutrition twins

In nutrition, Gkouskou et al. extended the idea of an "actual digital twin", in which a patient is matched to many similar patients whose outcomes are known, into a "virtual digital twin": a personal model combining genetic, longitudinal metabolomic, immune, microbial and behavioural data, collected through connected devices and managed by machine learning, to simulate dietary interventions for that person [**gkouskou2020virtual**]. Metabolic twins have also been outlined as tools that track a person's metabolic flexibility over time [**su2025metabolic**]. A systematic review of twins in precision health identified 12 studies and found improvements in most of the reported outcomes, while noting that the evidence is still small and heterogeneous [**shen2024effectiveness**].

The most developed clinical example is the Twin Health platform. Its method paper describes a machine-learning model (CatBoost) that predicts each person's post-meal glucose response from continuous glucose monitoring, food logs, activity and sleep, and a mobile application that adjusts dietary recommendations from those predictions and keeps refining them as data arrive [**twinhealth2024frontiers**]. In a one-year randomised trial in 319 people with type 2 diabetes, nutrition guided by this twin improved liver-fat scores and MRI-PDFF compared with standard care [**joshi2023digitaltwin**]. A second example, iDT-diet, modelled how diet quality affects biomarkers over time in 402 participants measured at baseline and after one, three and six years. It combined a random forest with Choquet-integral LASSO feature selection, which had lower error than XGBoost, LightGBM and CatBoost for several targets, with generated natural-language descriptions and three-dimensional avatars of the predicted body state [**nobel2026idtdiet**]. In MASLD specifically, the European ARTEMIS project is building virtual twins from retrospective data on several thousand patients [**artemis2024**].

These systems either recommend individual foods from predicted glucose responses or forecast biomarker trajectories. The Twin Health platform depends on continuous glucose monitoring and proprietary data, and iDT-diet forecasts biomarkers without recommending food. None evaluates complete, existing recipes against evidence-graded dietary priorities, and none combines health with cuisine and practicality when making a recommendation.

2.5.3 From prediction to "what if"

A twin that compares interventions must answer counterfactual questions, which predictive accuracy alone cannot license. Freeman et al. name two traps. The *fidelity trap* is the belief that a model accurate enough to reproduce patient trajectories can therefore say what would happen under a different treatment; prediction and counterfactual estimation are distinct tasks with different justifications. The *feedback trap* arises when a twin learns from data that its own recommendations helped to produce, which breaks the assumptions behind its causal estimates. They argue that health twins should be causally valid, modular and evolving systems, with modularity isolating the parts that need interventional justification [**freeman2026traps**]. According to Laudy, a digital twin should represent how an individual might respond under treatment and control scenarios. To assess its reliability, the framework defines three dimensions of fidelity—marginal, joint, and structural—and

outlines five levels of validation through which model predictions can be examined using observable data. However, Laudy acknowledges that the relationship between an individual's potential outcomes under the two conditions cannot be directly verified [**laudy2026counterfactual**].

For HepatoTwin, these arguments have a direct consequence. A cross-sectional model trained on observational data, such as a steatosis classifier built on NHANES, describes the current state well but cannot by itself predict the effect of changing a diet. Intervention effects must instead come from interventional evidence, held in a separate module with their strength and uncertainty made explicit, and the twin's updating must be evaluated in a setting where the true effect is known (Section 2.7).

2.6 Computational Gastronomy and Food Recommendation

2.6.1 Recipe and food resources

Computational gastronomy treats food and cooking as structured data, studying recipes through their flavour, nutrition, health associations and sustainability [**bagler2024gastronomy**]. RecipeDB contains 118,171 recipes from six continents, 26 geocultural regions and 74 countries, cooked with 268 processes from over 20,000 ingredients, with cuisine labels, utensils and nutritional profiles, and ingredients linked to their flavour molecules [**batra2020recipedb**]. FlavorDB catalogues 25,595 flavour molecules, 2,254 of which are linked to 936 natural ingredients in 34 categories [**garg2018flavordb**]. Literature mining of 5,769 abstracts produced 8,957 associations between 152 culinary spices and 848 diseases; for 150 of the 152 spices the beneficial associations outnumbered the adverse ones, and the authors used this balance to make disease-specific spice recommendations [**rakhi2018spices**]. These resources make it possible to reason about recipes at scale, but they are built for exploration and analysis rather than for deciding which recipes suit a particular patient.

2.6.2 Food recommender systems

Food recommendation faces challenges beyond those of other domains: it must balance taste, health, culture and practicality. In a survey of the field, Trattner and Elsweiler showed that re-weighting standard recommender outputs by WHO or UK Food Standards Agency health scores can trade accuracy for healthiness, but the healthiest lists came with very poor accuracy, and they named health-aware recommendation an open problem [**trattner2017food**].

Knowledge graphs are a common foundation. FoodKG links over a million recipes, their ingredients and USDA nutrient data in 67 million triples, and supports queries such as which recipe can be made from the ingredients at hand while avoiding allergens [**haussmann2019foodkg**]. Chen et al. framed personalised recommendation as constrained question answering over FoodKG, treating the user's query, dietary preferences and health guidelines as constraints, and improved on non-personalised baselines by 59.7 percentage points on average across metrics [**chen2021constrained**]. Graph neural networks have been used to move users gradually towards healthier recipes while

respecting their preferences [**li2022healthguided**]. In all of these systems, "healthy" is defined by generic nutritional guidelines rather than by a patient's clinical state, and the evidence behind each recommendation is not graded.

2.6.3 Recipe modification and adaptation

A related line of work modifies recipes by substituting ingredients. Shirai et al. combined explicit knowledge from a food knowledge graph with word embeddings into a substitutability heuristic (DIISH), filtered the candidates by nutritional and food-category constraints, and evaluated it against substitutions taken from cooking guides and recipe reviews [**shirai2021substitutions**]. Loesch et al. learned knowledge-graph embeddings (RDF2Vec) over a food graph and evaluated them against 1,847 known substitution pairs, retrieving about 40% of the true substitutes in the top ten candidates [**loesch2022substitution**]. Rita et al. fine-tuned large language models to choose substitutions that raise phytochemical content, improving Hit@1 from 34.5% to 38.0% on the GISMo benchmark and producing 1,639 enriched recipes, while cautioning that the health claims rest on preclinical evidence [**rita2024phytochemical**]. A survey of the area concludes that a good substitute must match flavour, culinary function and health suitability together, and recommends knowledge graphs and neuro-symbolic reasoning for doing so [**kim2025substitution**]. Cultural adaptation goes further: translating recipes between Chinese and English cuisines requires changing ingredients, units, tools and techniques, and GPT-4 still fell short of human experts when adapting English recipes into Chinese [**cao2024cultural**].

Every modification creates a new recipe whose taste, feasibility and nutrition must be predicted, and none of these studies tested whether people liked or cooked the result. Evaluating existing recipes, the approach taken in this thesis, avoids that uncertainty: every recommended recipe has already been written, cooked and nutritionally described.

2.6.4 Menu planning, diversity and additional criteria

Menu planning composes meals that satisfy nutritional and cost targets; Sahin et al. released EvoMeal, an open-source tool comparing four multi-objective evolutionary algorithms, and found that two newer algorithms (AGE-MOEA and SMS-EMOA) outperformed NSGA-II and NSGA-III [**sahin2024menu**]. Recommendation lists also benefit from diversity: maximal marginal relevance re-ranks items by a linear combination of relevance and dissimilarity to the items already chosen, also reducing redundancy [**carbonell1998mmr**]. Two further criteria are relevant to MASLD guidance. The degree of food processing can be estimated with FPro, a machine-learning score that predicts processing from nutrient profiles; it classified over 73% of the US food supply as ultra-processed, was associated in NHANES with metabolic syndrome and diabetes, and showed that swapping foods for less-processed alternatives could reduce these associations [**menichetti2023processing**]. Glycaemic load is available from the international glycaemic index tables [**atkinson2021gi**].

2.7 Evaluating Digital Twins Without Longitudinal Data

A twin's defining features, updating with new data and projecting the future, can only be fully validated against repeated measurements of the same people. When such data are unavailable, simulation offers a recognised alternative. In-silico clinical trials test interventions on computer-simulated patients before, or alongside, real trials [**pappalardo2019insilico**]. The best-known example is the UVA/Padova type 1 diabetes simulator, a population of virtual patients built from physiological models, which the US Food and Drug Administration accepted in 2008 as a substitute for animal trials in the preclinical testing of closed-loop insulin control [**kovatchev2009insilico**, **dallaman2014simulator**]. Its authors stress that such testing is a prerequisite to clinical trials, not a substitute for them [**kovatchev2009insilico**]. Synthea generates synthetic patient records for software testing at scale [**walonoski2018synthea**].

Two designs are particularly relevant. In plasmode simulation, real data provide the covariates and the outcome is simulated from a known model, so the data keep their real structure while the true effect is known and a method's ability to recover it can be measured [**franklin2014plasmode**]. In data assimilation, observing system simulation experiments generate a "true" state with one model, give a forecasting system only noisy observations of it, and measure how well the system recovers the truth; the field calls these twin experiments [**kalnay2003assimilation**].

Simulated follow-up is credible only if its ingredients come from evidence. For diet-driven weight change, Hall et al. developed and validated a dynamic model of how a sustained change in energy intake translates into weight over time: a lasting reduction of about 10 kcal per day eventually lowers weight by about 0.45 kg, with half of the change reached in about one year and 95% in about three years. The common static rule of 3,500 kcal per pound overestimates long-term weight loss by about 100%, and the model predicts average rather than individual responses [**hall2011energy**]. Large lifestyle trials report the weight trajectories that real participants achieve. In Look AHEAD, intensive lifestyle intervention produced a mean weight loss of 8.6% at one year against 0.6% with usual care; 68.0% and 37.7% of participants lost at least 5% and 10%, and at eight years the mean losses were 4.7% and 2.1% [**lookahead2014eightyear**]. The Diabetes Prevention Program reported a similar pattern of loss and partial regain [**dpp2002**], and a Look AHEAD analysis showed that losses of 5–10% increased the odds of clinically meaningful improvements in glycaemia, blood pressure and lipids [**wing2011modest**]. The response of the liver itself is given by the weight-loss studies of Section 2.1 [**koutoukidis2019weightloss**, **vilargomez2015weight**, **lim2011reversal**]. Realistic observations also require the normal biological variation of each laboratory test [**eflm˙bv**] and the known under-reporting of food intake [**subar2003open**].

These sources make it possible to evaluate a twin's updating on virtual patients whose follow-up is grounded in published human data. Such an evaluation verifies that the twin behaves correctly when the truth is known; it does not establish its accuracy for real patients, which still requires longitudinal data.

2.8 Summary and Research Gap

Table 2.3 summarises what each strand of the literature offers and what it leaves open. Clinical studies have established links between dietary changes, weight management, and liver health. However, their findings are generally reported in terms of energy intake, nutrient composition, and dietary patterns rather than individual recipes. Machine-learning models have shown promise in identifying hepatic steatosis using routinely collected clinical data, yet most are designed for one-time classification, with limited calibration and constraints on continuous prediction arising from genetic variability and measurement uncertainty.

Existing digital twins for nutrition support personalised food choices or predict changes in biomarkers, but they do not assess complete recipes against priorities ranked by the strength of clinical evidence. Moreover, causal inference research shows that a model's ability to predict outcomes accurately does not, by itself, demonstrate that its recommended interventions will lead to the expected results. Food recommendation and ingredient substitution systems can account for user preferences and general measures of nutritional quality, but typically lack integration with individual clinical conditions and the certainty of supporting evidence. In addition, the performance of liver-specific digital twins has not yet been adequately assessed in terms of how their predictions and recommendations evolve as new patient data become available.

Table 2.3: Strands of related work, what they leave open, and how HepatoTwin addresses it.

Strand	What it provides	What it leaves open	HepatoTwin
Clinical evidence (Section 2.1)	Graded effects of weight loss, diet patterns and nutrients	Not expressed for recipes; evidence strength varies	Evidence engine with explicit strength labels, mapped to recipe criteria
Steatosis models (Section 2.3)	Good discrimination from routine data	Snapshots; rarely calibrated; bounded R^2	Calibrated probabilities as the twin's state, within a defined scope
Dietary assessment (Section 2.4)	Recall data; usual-intake methods	Single-day recalls used directly	Recall-adjusted priorities; usual intake updated from food logs
Digital twins (Section 2.5)	Personal state, glucose-response prediction, trajectories	No recipe-level, evidence-graded recommendation; fidelity and feedback traps	Modular twin linking state, evidence and recipes; effects from interventional evidence
Food recommendation (Section 2.6)	Recipe data, preferences, generic healthiness, substitution	No clinical state; ungraded evidence; untested modified recipes	Evidence-gated ranking of existing recipes by health, cuisine and feasibility
In-silico evaluation (Section 2.7)	Virtual patients, plasmode and twin experiments	Not yet applied to liver twins	In-silico evaluation of the twin's updating on virtual NHANES patients

This thesis addresses the intersection of these strands: a patient-specific liver twin whose calibrated state, combined with graded intervention evidence and the patient's own diet, drives the transparent evaluation and ranking of existing recipes, and whose updating is evaluated in silico before real longitudinal data exist.

Chapter 3

Methodology

Chapter 4

Results